SPORTS & ARTHROSCOPY · NEET-SS / FET / INI-SS

Patellofemoral instability

1. Anatomy and biomechanics

- **Patella:** the **largest sesamoid**; **thickest articular cartilage in the body (up to about 5-7 mm)**; **lateral facet larger** than the medial; **odd facet** (medial, contacts in deep flexion).
- **Function: increases the moment arm of the quadriceps (by about 30-50%)**, centralises forces, protects the knee.
- **Contact areas:** at **20 degrees**, the **distal pole** contacts the trochlea; contact moves **proximally** on the patella with increasing flexion; **maximum contact area at about 90 degrees**; **at about 120-135 degrees**, the odd facet contacts the medial femoral condyle.
- **Patellofemoral joint reaction force: about 0.5 x body weight walking**, **3-4 x stairs**, **7-8 x squatting**; increases with flexion (and with open-chain extension near full extension: high quadriceps force).
- **Engagement:** the patella is **laterally positioned in extension** and **engages the trochlear groove at about 20-30 degrees of flexion**: **instability occurs in early flexion (0-30 degrees)**, where the **soft-tissue restraints matter most**.

Restraints to lateral patellar translation

- **Medial patellofemoral ligament (MPFL): primary restraint (about 50-60%) at 0-30 degrees**. **Femoral attachment (Schöttle point): between the adductor tubercle and the medial epicondyle** (just anterior to the posterior cortex extension line, proximal to the posterior origin of the medial condyle on a true lateral radiograph); **patellar attachment: proximal two-thirds of the medial border**. **Injured in almost all (about 90-100%) first-time dislocations**,

commonly at the **femoral attachment** (adults) or **patellar attachment** (children).

- **Medial patellotibial ligament, medial patellomeniscal ligament, medial retinaculum** (secondary, more important in deeper flexion), **vastus medialis obliquus (VMO)** (dynamic).
- **Trochlear geometry**: the main restraint **beyond about 30 degrees** of flexion.

2. Risk factors for instability (Dejour "major instability factors")

1. **Trochlear dysplasia** (**most important; about 85-96% of patients with recurrent instability**).
2. **Patella alta** (**delays engagement of the patella in the trochlea**).
3. **Increased tibial tubercle-trochlear groove (TT-TG) distance** (**over 20 mm abnormal**; normal about 10-13 mm; lateralised tubercle).
4. **Patellar tilt** (over 20 degrees on CT).
5. **Other factors: generalised ligamentous laxity (Beighton score)**, **femoral anteversion** (increased), **external tibial torsion**, **genu valgum (increased Q angle)**, **VMO dysplasia**, **female sex**, **young age (adolescents)**, **previous dislocation**, family history.
6. **Q angle:** line from ASIS to the patellar centre and from the patellar centre to the tibial tubercle: **normal about 14 degrees (men) and about 17 degrees (women)**; measured in extension (and in 30 degrees flexion to seat the patella).

3. Imaging and measurements

Radiographs

- **True lateral (posterior condyles superimposed):**
  - **Crossing sign** (**trochlear floor line crosses the anterior border of the lateral condyle**: **trochlear dysplasia**: the trochlea becomes flat at that point),

  - **Supratrochlear spur (bump)**,
  - **Double contour** (hypoplastic medial facet).
  - **Patellar height indices:**
  - **Insall-Salvati ratio** (**patellar tendon length / patellar length**; **normal 0.8-1.2**; **over 1.2 = alta**, under 0.8 = baja),
  - **Caton-Deschamps index** (distance from the **inferior pole of the patellar articular surface to the anterosuperior tibial plateau / articular surface length**; **normal 0.6-1.2**; **over 1.2 = alta**; **not affected by tibial tubercle osteotomy**: preferred for planning distalisation),
  - **Blackburne-Peel ratio** (perpendicular distance from the tibial plateau line to the inferior articular pole / articular length; **over 1.0 = alta**).
- **Axial (Merchant view, 45 degrees knee flexion): sulcus angle (normal about 138 ± 6 degrees; over 145 = dysplasia)**, **congruence angle (normal -6 ± 11 degrees; over +16 abnormal: lateral subluxation)**, **lateral patellofemoral angle** (Laurin: should open laterally).

CT/MRI

- **TT-TG distance** (**over 20 mm = abnormal**; MRI values are smaller than CT by about 2-4 mm), **TT-PCL distance (over 24 mm)** (less affected by trochlear dysplasia/knee rotation).
- **Trochlear dysplasia (Dejour classification):**
  - **Type A: crossing sign**; shallow trochlea (sulcus angle over 145 degrees) but preserved morphology,
  - **Type B: crossing sign + supratrochlear spur**; flat or convex trochlea,
  - **Type C: crossing sign + double contour**; asymmetric facets (convex lateral, **hypoplastic medial facet**),
  - **Type D: crossing sign + spur + double contour**; **"cliff" pattern** (vertical step between facets).

- **Patellar tilt**, **femoral anteversion** and **tibial torsion** (rotational profile CT), **MRI**: **MPFL tear**, **bone bruise** (**medial patellar facet and lateral femoral condyle**: pathognomonic of transient dislocation), **osteochondral fractures/loose bodies**.

4. Acute first-time patellar dislocation

- **Mechanism: twisting on a flexed, valgus knee with the foot planted** (internal rotation of the femur on a fixed tibia), or direct blow; usually **lateral** dislocation; often reduces spontaneously with knee extension.
- **Examination:** haemarthrosis, **tenderness over the medial retinaculum/MPFL femoral attachment (Bassett sign)**, **apprehension sign (Fairbank)**, **patellar glide (quadrants: over 2-3 quadrants lateral translation = laxity)**.
- **Osteochondral fractures** in about **25-50%** (**inferomedial patellar facet** (commonest) or **lateral femoral condyle**): radiographs/MRI.
- **Management:**
  - **Non-operative for most first-time dislocations** without loose bodies: **brief immobilisation for comfort**, **early ROM, quadriceps (VMO) and hip abductor/external rotator strengthening**, **patellar stabilising brace/taping**.
  - **Surgery after the first dislocation if: osteochondral fracture/loose body** (**fix or remove: arthroscopy**), **substantial MPFL disruption with lateral patellar subluxation on the axial view (relative)**, **high-risk patients** (young with **dysplasia + patella alta + open physes: recurrence risk over 50-70%**).
  - **Recurrence after the first dislocation: about 15-45% overall** (**up to about 70%** in skeletally immature with dysplasia).

5. Recurrent patellar instability: surgical options

- **Principle: address the predominant anatomical abnormalities ("à la carte" surgery, Dejour).**

MPFL reconstruction

- **Workhorse procedure** for recurrent instability: **gracilis or semitendinosus autograft**; femoral fixation at the **anatomic Schöttle point** (**fluoroscopic confirmation**); **graft fixed at about 30-60 degrees of flexion** with the patella centred in the trochlea; **avoid over-tensioning** (**medial patellofemoral overload, pain, loss of flexion, medial instability**).
- **Femoral tunnel malposition: too proximal** (graft **tight in flexion**: **loss of flexion and medial patellar overload**), **too distal** (tight in extension/loose in flexion).
- **Paediatric: femoral attachment is just distal to the distal femoral physis**: drill below the physis or use **physeal-sparing (soft-tissue/adductor sling) techniques**.
- **Complications: patellar fracture** (transverse tunnels across the patella: use **anchors or short sockets**), recurrent instability, stiffness, overconstraint.
- **MPFL repair** has higher recurrence than reconstruction.

Tibial tubercle osteotomy (TTO)

- **Medialisation (Elmslie-Trillat)**: **increased TT-TG (over 20 mm)**: **aim for TT-TG about 10-15 mm**; **avoid over-medialisation** (increases medial tibiofemoral and **medial patellofemoral** contact pressures).
- **Anteromedialisation (Fulkerson)**: **oblique osteotomy**: medialises + **anteriorises (unloads the distal and lateral patellar facets)**: **lateral facet or distal patellar chondral lesions** with instability/malalignment; **contraindicated for proximal/medial facet lesions** (shifts load onto them).
- **Distalisation**: **patella alta (Caton-Deschamps over about 1.2-1.3)**: target about 1.0.
- **Complications: proximal tibial fracture** (**avoid early full weight bearing with Fulkerson: risk of tibial fracture especially at 6-12 weeks**), non-union, **compartment syndrome**, **DVT**, **anterior tibial**

artery/deep peroneal nerve injury (posterior cortex drilling), painful hardware, **Hauser procedure (posteromedialisation: increases PFJ pressure: abandoned; causes PFJ arthritis)**.

- **Roux-Goldthwait** (lateral half of the patellar tendon transferred medially) in skeletally immature (avoids the tibial tubercle apophysis: **TTO contraindicated with open physes: genu recurvatum**).

Trochleoplasty

- **Severe trochlear dysplasia (Dejour B and D with a supratrochlear spur over 5 mm)**, especially **revision/failed MPFL reconstruction**: **sulcus-deepening trochleoplasty (Dejour/Bereiter)**, **recession wedge trochleoplasty** (for spur). **Complications:** arthrofibrosis, cartilage necrosis, PFJ arthritis; **contraindicated with open physes and established PFJ arthritis**.

Lateral retinacular release

- **Never as an isolated procedure for instability** (no benefit, can cause **iatrogenic medial patellar instability**). **Lateral lengthening** for a **tight lateral retinaculum with a positive patellar tilt test (tilt under 0 degrees: cannot lift lateral facet to neutral)**.

Rotational/angular osteotomies

- **Femoral derotational osteotomy** for **excessive femoral anteversion (over about 25-30 degrees)**; **distal femoral varus osteotomy** for **genu valgum** (over about 5 degrees mechanical valgus) in recurrent instability; **tibial derotation** for external tibial torsion.

6. Special situations

- **Habitual (obligatory) dislocation:** dislocates with **every flexion**, in children: **quadriceps contracture** (**vastus lateralis/iliotibial band contracture**; prior quadriceps injections): **quadricepsplasty/lengthening with realignment**.

- **Congenital (permanent) dislocation:** irreducible from birth; associated with **Down syndrome, nail-patella syndrome (LMX1B: hypoplastic patella, iliac horns, renal disease)**, **arthrogryposis**: early soft-tissue realignment.
- **Medial patellar instability: iatrogenic after lateral release** (vastus lateralis tendon detachment); **lateral retinaculum reconstruction/repair**.
- **Patellofemoral pain syndrome (anterior knee pain): diagnosis of exclusion**; **hip abductor/external rotator and quadriceps strengthening** (best evidence), taping, orthoses; **no surgery**.
- **Patellar tendinopathy (jumper's knee): eccentric/heavy slow resistance exercise**, **decline squats**; **Blazina staging**; surgery (debridement) for refractory cases.
- **Sinding-Larsen-Johansson** (inferior pole traction apophysitis) and **Osgood-Schlatter** (tibial tubercle traction apophysitis): self-limiting; **excise the ossicle** after maturity if symptomatic.

7. Summary: planning algorithm

Pathology	Threshold	Procedure
Torn/insufficient MPFL	Recurrent instability	**MPFL reconstruction (all cases)**
TT-TG increased	**Over 20 mm**	**Medialising TTO** (± anteriorisation for lateral/distal facet wear)
Patella alta	**Caton-Deschamps over 1.2**	**Distalising TTO** (or MPFL alone if mild)
Severe trochlear dysplasia	**Dejour B/D, spur over 5 mm**	**Trochleoplasty** (revision mainly)
Femoral anteversion	**Over about 25-30 degrees**	**Femoral derotation osteotomy**

Pathology	Threshold	Procedure
Valgus	Mechanical valgus over about 5 degrees	**Distal femoral varus osteotomy**
Tight lateral retinaculum	Tilt test negative	**Lateral lengthening** (never isolated release)

Exam pearl: **MPFL = primary restraint 0-30 degrees; femoral attachment (Schöttle point) between the adductor tubercle and medial epicondyle; torn in nearly all dislocations. Dejour major factors: trochlear dysplasia (crossing sign), patella alta, TT-TG over 20 mm, tilt over 20 degrees.**

Exam pearl: **Bone bruise on the medial patellar facet and lateral femoral condyle = transient patellar dislocation. Osteochondral fracture of the inferomedial patella: fix or remove. First-time dislocation without loose body: non-operative.**

Exam pearl: **Proximal femoral tunnel in MPFLR = tight in flexion, medial overload. Fulkerson AMZ unloads lateral/distal facets (avoid for proximal/medial lesions). Caton-Deschamps index is used for distalisation. Never isolated lateral release (iatrogenic medial instability). TTO contraindicated with open physes (recurvatum): Roux-Goldthwait.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.